FIT3155 SAMPLE EXAM PAPER

Generated Sample Exam 03

INSTRUCTIONS

- This exam is worth 60 marks.

General exam technique

- Some students lose marks by not attempting all questions.
- Suppose you are likely to get 3/6 on a question after spending say 10 minutes on it. Spending another 10 minutes on the same question may gain you at most 3 additional marks. In contrast, spending those 10 minutes on a new question could earn you another 6 marks. The key point is: do not get stuck on one question.
- A good strategy is to first answer the questions you are most confident about before attempting the others. In other words, answering questions strictly serially during an exam can be suboptimal.
- Do not waste time answering an unrelated question in the hope of gaining partial marks. Marking will be based on the question that was actually asked.
- Where necessary, justify your answers with clear and concise explanations, without being overly wordy.

Question 1 **(6 marks)**

Let $S[1 \ldots m]$ and $T[1 \ldots n]$ be two strings. An *overlap* of S into T is an integer ℓ such that the suffix $S[m - \ell + 1 \ldots m]$ is exactly equal to the prefix $T[1 \ldots \ell]$.

Write pseudocode for an efficient algorithm that accepts S and T and outputs the maximum overlap length ℓ.

For this problem, you may assume that `zalg` is available as a built-in function that takes a string as input and returns the array of its Z-values. Do not write pseudocode for `zalg`. State the worst-case time and space complexity of your algorithm.

6

(Page intentionally left blank for working space)

Question 2 **(6 marks)**

In the Boyer–Moore algorithm, the extended bad-character rule stores $R_k(x)$, the position of the rightmost occurrence of character x strictly to the left of position k in the pattern $pat[1 \dots m]$, or 0 if no such occurrence exists.

(i) Write pseudocode to construct the full $m \times |\Sigma|$ table of $R_k(x)$ values for a fixed alphabet Σ.

(ii) Suppose a right-to-left scan mismatches at pattern position k against text character x. Explain how the extended bad-character shift is computed from $R_k(x)$, and state the preprocessing time and space complexity.

6

(Page intentionally left blank for working space)

Question 3 **(6=2+4 marks)**

Assume that \$ is a unique terminal symbol that is lexicographically smaller than every other character. The string $S =$ `banana$` has

$$L = BWT(S) = \texttt{annb\$aa}$$

and suffix array

$$SA = [7, 6, 4, 2, 1, 5, 3].$$

Using L as a Burrows–Wheeler search index, search for the pattern $pat =$ `ana` by backward search.

(i) Construct the needed $rank(c)$ values and the occurrence counts needed during the search.

(ii) Show the values of sp and ep after each processed pattern character. Then report the multiplicity of pat in S and the corresponding starting positions using the suffix array.

6

(Page intentionally left blank for working space)

Question 4 **(6=3+3 marks)**

These questions relate to Ukkonen's linear-time suffix tree construction algorithm. Let I_n be the implicit suffix tree for a string $S[1 \ldots n]$ that does not contain the terminal character \$.

(i) In the final phase that appends the unique terminal character \$, can any explicit suffix extension be completed by Rule 3? Answer Yes or No and justify your answer.

(ii) Explain why appending \$ converts the final implicit suffix tree into a regular suffix tree with one leaf for every suffix. In your answer, connect this to the role of Rule 2 and the rapid leaf-extension trick.

6

(Page intentionally left blank for working space)

Question 5 **(6=3+3 marks)**

(i) Decode the following Elias–Omega codeword to recover the positive integer N that it encodes:

`00000010100`

Show each component read during decoding.

(ii) The following tuples were produced by the LZ77 method:

$$\langle 0, 0, m\rangle, \quad \langle 0, 0, i\rangle, \quad \langle 0, 0, s\rangle, \quad \langle 3, 3, p\rangle, \quad \langle 4, 4, !\rangle.$$

Decode these tuples to regenerate the original string.

6

(Page intentionally left blank for working space)

Question 6 **(6 marks)**

Perform one iteration of the Miller–Rabin primality test on $n = 91$ using base $a = 3$.

Your answer should:

- express $n - 1$ in the form $2^s t$, where t is odd;
- compute the sequence of values checked by this Miller–Rabin iteration;
- state whether this base proves that 91 is composite, or whether this iteration is inconclusive.

6

(Page intentionally left blank for working space)

Question 7 **(6 marks)**

In a Fibonacci heap, after Extract-Min removes $H.min$, the children of the removed root are promoted into the root list. The consolidation step then repeatedly links roots of equal degree until no two roots in the root list have the same degree.

Write pseudocode for this consolidation step. Your pseudocode should use an auxiliary array indexed by degree and should link two roots of equal degree by making the larger-key root a child of the smaller-key root.

State the worst-case actual time and amortized time of Extract-Min.

6

(Page intentionally left blank for working space)

Question 8 **(6 marks)**

A dynamic array starts empty. The first append creates capacity 1. Whenever an append is attempted on a full array, a new array with twice the previous capacity is allocated, all existing elements are copied into it, and then the new element is appended.

Using the potential method, determine the amortized time complexity of an append operation over a sequence of n appends. Use the potential function

$$\Phi(A_i) = 2 \cdot size(A_i) - capacity(A_i).$$

Verify the required boundary conditions for Φ, and show the amortized cost calculation for both a non-resizing append and a resizing append.

6

(Page intentionally left blank for working space)

Question 9 **(6 marks)**

Let P be a node in a B-tree of minimum degree t. The node stores sorted keys $key_1, \ldots, key_r$ and, if it is not a leaf, child pointers $child_0, \ldots, child_r$.

Write pseudocode for a function $\textsc{Search}(P, x)$ that returns whether key x occurs in the subtree rooted at P. Your pseudocode should use binary search inside each visited node before deciding which child subtree to search.

Justify that your procedure terminates correctly, and state its worst-case time complexity in terms of the B-tree height h and minimum degree t.

6

(Page intentionally left blank for working space)

Question 10 **(6 marks)**

Use the Hungarian algorithm to solve the following minimum-cost assignment problem.

	Task 1	Task 2	Task 3
Agent 1	9	2	7
Agent 2	6	4	3
Agent 3	5	8	1

Show the row and column reductions you perform, identify a minimum-cost perfect matching, and state the total assignment cost.

6

(Page intentionally left blank for working space)

End of Exam